

NEURAL BELLMAN OPERATORS

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Neural Bellman Operators learn a common continuation value for repeated economic decisions. We characterize its reuse and assess nonlinear continuation contrasts with an independent statistic that cancels common simulation noise. In a nonlinear capital economy, two previously unpiloted fifty-dimensional decision menus establish material NBO payoff gains over materialized simulation and direct-policy actors. Eight of 24 conditional risk comparisons favor NBO, four favor simulation, and twelve remain unresolved. Exact closure of simultaneous payoff intervals bounds accuracy across every declared method and budget. In the fifty-dimensional Intermediate design, NBO is the least-cost candidate certified within 10^{-4} of the best catalogue payoff; cheaper uncertified alternatives remain unresolved. Independent paired confirmation, exact-trace neural-HJB comparisons, signed Bellman gains, and separate scalar and diffusion accounts retain their distinct economic targets. Complete applications preserve recursive utility, endogenous preferences, temporal selves, and strategic interaction.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

An economic decision often requires a continuation value that must itself be computed. A household changes consumption and portfolio risk while investing in its preferences. A sophisticated consumer anticipates future selves, and a firm invests against the policies of its rivals. These problems differ in their utility domains and deviation criteria, but each requires an evaluation of the consequences that follow a proposed action. The preference-formation problem of Qi (2025) provides a motivating example in which internal and external choices interact.

This paper develops Neural Bellman Operators (NBO), an evaluation and improvement map for these economic problems. A critic estimates the continuation value of a specified policy. A feasible actor uses its derivative channels to change current decisions, keeping the evaluated continuation fixed. The resulting policy is assessed against an explicit economic criterion. Boundary conditions, information, numerical approximation, and the deployed decision rule are part of that criterion.

The central question is when learning a continuation value makes economic computation more effective. A fitted continuation can be reused across states and action queries, but fitting is costly and its approximation error can alter the selected action. A credible comparison must allow a simulation-based method to cache the information that the same decision problem permits. It must also include the cost of training and choosing the continuation representation. We therefore study the accuracy and complete work of the learned evaluation block when several economic decisions use a common continuation. The nonlinear capital economy provides the common primitives for this comparison and for the policy-specific error analysis.

The theory follows the evaluation–decision–payoff chain. A fixed-policy equation identifies the continuation being approximated. The improvement map specifies the actor’s feasible action and the derivatives it can use. General verification results distinguish an evaluation defect from unfinished action maximization. For the capital economy, an exact finite Bellman identity places the relevant continuation error on the action change under the candidate’s own occupation law. A signed continuation calculation can assess the same identity using common future paths, retaining the cancellation between predicted improvement and prediction error. Its payoff range depends on the economy and action tube. The continuous-time account separately charges the implementation, diffusion, clipping, and numerical errors of the held controller. These are error bounds for computed objects; they do not assert a convergence rate for a neural optimizer.

The computational design uses two complementary forms of comparison. A repeated-decision study examines the incremental value of reusing a learned continuation, including feasible Raw caching and the complete cost of fitting and selection. Independent confirmation on the original capital policies examines the previously reported paired payoff comparisons using a proof, constants, reporting code, and

confidence allocation fixed before the new simulation bank. A strengthened neural-HJB implementation uses an exact diffusion trace, independent residual selection, and a documented action implementation. A separate signed Bellman experiment assesses the precision of the occupation account. The design and reporting preserve the distinction between new decision problems, new payoff observations on existing policies, and changes in the analytical treatment of existing observations.

The continuation menu gives a direct economic result. In the two previously unpiloted fifty-dimensional Long and Intermediate designs, NBO’s protected payoff gains over both the materialized Raw actor and direct policy exceed the prespecified margin 10^{-4} . Its complete construction and query time is also lower than direct policy’s in those cells. At the final prescribed budget, six of eight cells have NBO-cached-SAA intervals wholly within $\pm 10^{-4}$, with lower construction and query work for NBO. Cached SAA delivers materially higher payoffs in the other two cells, and the cheaper earlier SAA budgets remain in the comparison. These results concern the complete sixteen-stream populations and 384 fixed task-state queries in each finite economy. They identify useful fitted-continuation decisions and their observed costs. A separate assessment below tests continuation prediction directly, and a simultaneous catalogue calculation compares every declared budget; neither establishes a unique causal mechanism or a minimum-work frontier over all possible procedures.

The independent original-policy confirmation also makes the paired comparison prospective: NBO exceeds the original neural-HJB implementation by 10^{-4} in both dimensions. A newly trained, strengthened HJB procedure gives a more demanding comparison across four continuous-diffusion economies. Only the low-volatility, fifty-dimensional cell establishes that margin against both its greedy and distilled deployments; the other direct comparisons remain unresolved. The signed Bellman assessment gives continuous-economy gain lower bounds above 0.0005 in both dimensions while its signed continuation-correction intervals still contain zero. The full adapted-class regret bounds remain distinct from these gains and from the finite menu’s scalar accuracy certificates.

The previous complete sixteen-stream experiment remains part of the current evidence. Its direct NBO–Raw and NBO–DPO payoff intervals were unresolved,

and cached Raw was cheaper at the fifty-dimensional online target. A common-innovation transfer refinement made the NBO–HJB comparison positive in both ten and fifty dimensions, but the refinement was developed after the original analytical freeze and inspection of interval widths. The original dimension-ten interval included zero. We retain both intervals and their chronology. The original mechanism lower bound was nonpositive in each dimension. Those observations motivate the current experiments and remain visible alongside them; improving a verifier does not by itself identify a benefit of learning the continuation.

Two further results connect this evidence to the learned object and the economic accuracy target. Independent prediction assessment compares the nonlinear scalar continuation with realized Raw caches at the same selected action pairs. Common future-path noise cancels from the difference of squared errors. Eight of 24 pre-specified intervals establish lower NBO contrast risk, four favor Raw, and twelve remain unresolved. This measures a conditional prediction benefit without inferring payoff ordering from risk ordering. Separately, exact closure of all 216 simultaneous menu payoff intervals bounds regret relative to the complete sixteen-candidate budget catalogue. In the fifty-dimensional Intermediate design, final-stage NBO is the least-cost candidate certified within 10^{-4} of the best catalogue payoff. Its mean complete construction and query cost is 12.6993 seconds, rounded upward. Cheaper uncertified candidates remain possible competitors, and the full cost of conducting the comparison remains charged. This last calculation is an explicitly post-freeze deterministic use of simultaneous intervals, not a new prospective stopping experiment.

The economic applications remain part of the paper’s substantive scope. Recursive utility evaluates each policy through its own utility process. Endogenous preference adjustment connects an internal shadow value with consumption and portfolio demand. Temporal selves require a local spike-deviation equilibrium, whereas strategic firms require a complete unilateral best response against fixed rivals. The main article states these economic distinctions and their principal implications. The current *Economic Applications* companion contains their full models, derivations, proof arguments, and numerical comparisons. The technical supplement contains the general theory, detailed capital accounts, and complete compu-

tational records. This organization places the common numerical question in the foreground while preserving the distinct economic problems that give the operator its content.

1.1. *Relation to the literature*

Neural Bellman Operators build on numerical dynamic programming and function approximation. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Smooth neural networks can approximate derivatives under representation hypotheses ([Hornik et al., 1990](#)). These results distinguish representational capacity from the accuracy achieved by a finite architecture and a particular optimizer.

The differential critic is related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. Our object combines a specified policy continuation, a feasible actor, and an economic error account, and asks when the continuation representation is useful across decision queries. An analytical maximizing action is useful when available; it is not a universal distinction between neural PDE methods. Recursive dependence on the value is a nonlinear zeroth-order term, whereas full nonlinearity of a PDE concerns its highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Utility-process selection and transversality matter in its infinite-horizon formulation ([Herdegen et al., 2021](#)). Temporal selves instead require an equilibrium formulation ([Björk et al., 2016](#)). The stochastic approximation literature provides convergence results under explicit step-size, stability, and limiting-dynamics conditions ([Borkar, 2024](#)). We retain those distinctions when describing a finite neural computation.

Section 2 defines the economic object and Section 4 specializes it to the capital economy. Section 4.3 connects continuation error with economic gain. The computational design in Section 5 distinguishes continuation reuse, conditional payoff replication, and mechanism assessment. Section 9 states the other economic appli-

cations. General verification results and their full proofs remain in the technical supplement, beginning with Section S.1; the application companion is a current part of the paper.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy*

values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q \sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this paper uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility,

integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T}|V(S_T)|] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section A.1.3.

3. NEURAL BELLMAN OPERATORS

The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_\xi(t, x) = G(x) + (T - t)N_\xi(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\xi} = G_B + d_B N_{\xi}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\xi} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\xi}] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_{\nu} \mathcal{H}^{a_{\phi}}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction is substantive: joint differentiation can move the critic away from the policy-evaluation root. Appendix S.3 gives a counterexample, the recursive lifetime policy derivative, and the conditioning requirements. The local Hamiltonian objective is not identified with a lifetime policy gradient under arbitrary state sampling. The evaluated reference policy may remain fixed while a separate candidate actor is improved at successive work levels. Updating that reference defines a new evaluation target and is recorded explicitly.

3.3. Rollout and costate evaluation

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\hat{V}^{(j)}(t, y)$ and its pathwise derivative $\hat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\hat{\mathbb{E}} \left[0.05 \{v_\theta - \hat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \hat{p}^{(j)})\|_d^2 \right]. \quad (8)$$

The factor d puts normalized marginal values on an order-one scale. The rollout implementation uses two hidden tanh layers and a hard terminal component. Holding its new derivative fixed, the actor then maximizes

$$\hat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (9)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. The number of critic and actor steps per rollout batch is a declared design parameter. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

3.4. *The implemented map and its verification output*

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections S.11.1 and A.1.3.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

3.5. *Evaluation, feasible improvement, and approximation*

The algorithm produces a continuation approximation and a feasible policy. Its economic interpretation requires an error account for each. The general results in technical Section S.1 bound the consequence of a fixed-policy evaluation error and an incomplete feasible maximization under the stated comparison and boundary conditions. The positive-weight Bellman formulation also provides a monotone

Algorithm 1 Neural Bellman evaluation, feasible improvement, and economic assessment

Require: Economic primitives and information; feasible actions and utility domain; boundary data; evaluation variant; training, validation and stopping rules; independent final assessment.

- 1: Declare the feasible reference policy to be evaluated. Initialize a feasible candidate and a boundary-compatible continuation representation.
 - 2: **for** each permitted evaluation-improvement iteration **do**
 - 3: Freeze the declared reference policy, rivals and evaluation target. Fit only the critic by the declared differential, positive-weight, or rollout rule. The reference may remain fixed across work levels.
 - 4: Freeze the critic and its value/derivative channels. Update only the feasible actor, or apply the declared feasible action solver.
 - 5: Record all evaluation and improvement work, corrections, unsuccessful checks, and the resulting deployment rule.
 - 6: Apply the predetermined validation and stopping rule, using no final-test outcomes.
 - 7: **end for**
 - 8: Freeze the selected policy and continuation object before independent final evaluation.
 - 9: Compute the specified economic endpoints and direct contrasts, with their approximation, implementation and statistical accounts.
 - 10: Return the fitted objects, their implementation and domain, and the complete accuracy and work record.
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approximation route when classical derivatives are unavailable. The complete statements and proofs remain current in the supplement.

The following capital economy permits a more operational account. Its payoff identity follows the implemented action through the original diffusion. Its finite Bellman identity measures continuation error along the candidate's actual action change. This specialization makes it possible to compare a learned continuation

with a cached simulation continuation using the same economic decision and a common assessment design.

4. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule. Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z}1$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - \nu^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (11)$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{\overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2\} dt + e^{-\rho T} \{\bar{Y}_T - \chi \|\Pi Y_T\|_d^2\} \right], \quad (12)$$

where $\nu^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\sigma_I = 0.15$, $\sigma_C = 0.10$, $\zeta = 0.2$, $\chi = 0.03$, and $[\underline{m}, M] = [0.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (13)$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. For the analytical comparison, keep m_0 interior and assume $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, and let $\beta \geq \kappa \|B\|_2$ and $s_0 \geq \|\Pi y\|_d$. Choose

$\epsilon \geq 0$ so that $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$ at every date. With $V^*(y) = \sup_m J_y(m)$ over the full adapted admissible class, the schedule gap obeys $0 \leq V^*(y) - J_y(m_0) \leq G_d$. The exact scalar integral defining G_d is (S.23); Appendix S.5 gives its curvature and dispersion constants, Theorem S.9, and the complete proof. This analytical bound supplies a common economic reference. The policy-specific endpoint below measures what the fitted actor adds to that reference.

4.1. The deployed neural map and its numerical guard

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (14)$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (15)$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its interval midpoint and is not evidence of successful training. This scalar feasibility guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

4.2. Policy-Specific Verification after Bellman Improvement

The bound in Theorem S.9 is a useful starting point, but not a measure of what training accomplishes. Every policy in the same tube receives the same allowance,

including a poorly trained policy. We now replace that allowance by an independently verified, policy-specific payoff difference. The economic primitives, initial state, and full adapted comparison class in (11)–(12) are unchanged. The implementation to which the new bound applies is specified explicitly below. It is not identified with continuously observed-state feedback.

4.2.1. The deployed controller

Fix a deterministic grid $t_k = kh$, $Nh = T$. The controller has an internal state Z_k , initially y , and observes the Brownian innovations at the grid dates. At date t_k it selects

$$a_k = \mathcal{P}_k\{m_\phi(t_k, Z_k)\}, \quad Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\Delta\tilde{W}_k, \quad (16)$$

where $c = a - v^2/2$, $f(z) = \kappa \tanh(Bz)$, and $\mathcal{P}_k$ projects inward onto the declared action interval around $m_0(t_k)$. The vector $\Delta\tilde{W}_k$ clips each standardized innovation at ten before rescaling by $\sqrt{h}$. The numerical implementation and its rounding convention form part of the controller. Between dates it holds a_k fixed. Its economic state, denoted Y^ϕ , nevertheless satisfies

$$dY_t^\phi = \{c\mathbf{1} + f(Y_t^\phi) - a_k\} dt + \Sigma dW_t, \quad t_k \leq t < t_{k+1}.$$

In particular, the economic Brownian motion and state are not clipped, stopped, or replaced by Z . The controller is adapted to the original innovation filtration and remains feasible. Observing innovations is an implementation assumption, not a conclusion inferred from a state-only observation model. The continuously evaluated absolute-bound study policies remain separate objects in the supplement.

This specification has an important analytical consequence. When the numerical state and the economic state are compared, they use exactly the same chosen actions. Their state discrepancy therefore depends on the smoothness of the economic drift, not on a global Lipschitz bound for the learned network. The following construction uses this fact rather than inserting sampled residuals into a uniform verification theorem.

4.2.2. An exact improvement identity

Write $W(t) = e^{-\rho t} w(t)$ and define the nonnegative consumption deficit

$$R_t(m) = \log m_0(t) - \overline{\log m} + w(t)\{\bar{m} - m_0(t)\} + \frac{\zeta}{2}\{\bar{m}^2 - m_0(t)^2\}. \quad (17)$$

The first-order identity $m_0^{-1} = w + \zeta m_0$ and concavity of the logarithm imply $R_t(m) \geq 0$. The exact mean-state identity in the proof of Theorem S.9 gives

$$\begin{aligned} \Delta_\phi := J_y(a^\phi) - J_y(m_0) &= \mathbb{E} \int_0^T W(t) \overline{f(Y_t^\phi) - f(Y_t^0)} dt - \mathbb{E} \int_0^T e^{-\rho t} R_t(a_t^\phi) dt \\ &\quad - \chi e^{-\rho T} \mathbb{E}\{\|\Pi Y_T^\phi\|_d^2 - \|\Pi Y_T^0\|_d^2\}. \end{aligned} \quad (18)$$

Thus state-dependent consumption has to earn its utility-curvature cost through production or terminal dispersion. The identity is also an effective variance reduction: the common mean-state martingale cancels before simulation statistics are formed. Positive neural output is not itself evidence of positive economic improvement.

For each cell, calculate the scalar integrals

$$\begin{aligned} A_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, & B_k &= \int_{t_k}^{t_{k+1}} W(t) dt, \\ C_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} \{\log m_0(t) - w(t)m_0(t) - \zeta m_0(t)^2/2\} dt. \end{aligned}$$

The consumption-deficit integral for a held action is exactly $C_k - A_k \overline{\log a_k} + B_k \bar{a}_k + \zeta A_k \bar{a}_k^2/2$. Consequently, consumption need not be approximated by a left-point utility sum. Let $X_{\phi,h}$ be the paired statistic obtained from (18) by using the internal Euler states in the production and terminal terms and these integrated consumption terms. The reference internal state uses the integrated common action $\int_{t_k}^{t_{k+1}} m_0(t) dt$. All scalar integral enclosures and their midpoint errors are saved.

4.2.3. Transferring the computed payoff to the economy

The statistic $X_{\phi,h}$ is evaluated for the held controller in (16), using the same actions when comparing its internal state with its economic diffusion. Proposition S.10 in Appendix S.5 bounds $|\Delta_\phi - \mathbb{E} X_{\phi,h}|$ by the saved allowance $E_{\phi,h}^{\text{all}}$. The exact-arithmetic diffusion allowance is $E_\epsilon(h) + E_0(h)$, with E_ϵ defined in (S.32).

The appendix gives the additional clipped-innovation, forward-arithmetic, and enclosed-integration terms. The $O(h)$ diffusion bound uses bounded actions, the global drift and derivative bounds of the specified hyperbolic-tangent production function, and additive diffusion; it requires no derivative bound for the trained network.

4.2.4. A post-training certificate

Use the deterministic clipping threshold b_ϕ and expectation-tail allowance τ_ϕ constructed in Appendix S.5.4. They account for the unbounded Gaussian terminal-dispersion term and are not estimated from the final sample range. Let $X_{\phi,h}^c$ be $X_{\phi,h}$ clipped to $[-b_\phi, b_\phi]$, so that $|\mathbb{E}X_{\phi,h} - \mathbb{E}X_{\phi,h}^c| \leq \tau_\phi$.

For n fresh independent paths, let $\bar{X}_\phi^c$ and s_ϕ^2 be the clipped empirical mean and unbiased variance. Fix a family of at most F one-sided statements before inspecting this noise bank, and set $\ell = \log(2F/\alpha)$. Define

$$L_\phi = \bar{X}_\phi^c - \sqrt{2s_\phi^2 \ell / n} - \frac{14b_\phi \ell}{3(n-1)} - E_{\phi,h}^{\text{all}} - \tau_\phi. \quad (19)$$

In the implementation the empirical mean is enclosed downward and the variance and every subtracted quantity upward. The upper payoff endpoint uses the same terms with reversed signs. The factor F counts both tails when both are reported.

THEOREM 1—Policy-specific continuous-time certificate: *Condition on the training and validation data, and suppose the specified final simulation paths are independent draws from the declared innovation law. Under the arithmetic contract and transfer assumptions above, with probability at least $1 - \alpha$, simultaneously for all declared policies and initial states,*

$$J_y(a^\phi) - J_y(m_0) \geq L_\phi, \quad 0 \leq V^*(y) - J_y(a^\phi) \leq G_d - L_\phi. \quad (20)$$

The optimum V^ still ranges over all adapted feasible controls in the original continuous economy, not merely the networks, schedules, or sampled controllers used in the experiment. Common shocks may be reused across policies; independence across those policies is not required. The guarantee is pointwise in each declared initial state and conditional on training, rather than uniform on an uncountable state domain.*

PROOF: Apply the empirical Bernstein inequality of [Maurer and Pontil \(2009, Theorem 4\)](#) to $(X_{\phi,h}^c + b_\phi)/(2b_\phi)$ and use a union bound over the declared one-sided statements. Rescaling gives the first two error terms in (19). The clipping expectation bound and Proposition S.10 then give the first inequality in (20). Subtract it from the deterministic anchor bound (S.23). Feasibility yields nonnegative regret. An observed negative upper regret endpoint would indicate failure of the confidence event or of an implementation assumption, not a negative optimality gap. *Q.E.D.*

Unlike $G_d + D_{d,\epsilon}$, the new endpoint depends on the deployed policy through its paired outcomes and sample variance. It can distinguish seeds and unsuccessful policies in the same architectural tube. When $L_\phi > 0$, the post-training guarantee is tighter than the analytical schedule’s guarantee. This is a statement about the verified policy, not a deterministic promise that every training run improves welfare. The concentration inequality, the identity that transfers a lower improvement bound into a regret bound, and conservative policy improvement are established ideas ([Kakade and Langford, 2002](#), [Pirodda et al., 2013](#), [Scherrer et al., 2015](#)). The contribution here is their operational connection to the continuous capital economy through (18) and the transfer bound (S.32), without a state-covering certificate or a global bound on the trained network’s derivatives.

4.3. An operational evaluation–improvement account

The costate in a rollout is the derivative of a specified continuation problem. Its approximation error has to be measured where the new policy uses that continuation. We now make both objects explicit. The construction uses the capital economy, including its dense production interaction, common and idiosyncratic shocks, consumption curvature, and terminal dispersion preference. It also accommodates a critic fitted on a coarser grid than the one used for assessment.

Fix the assessment grid $t_k = kh$, $Nh = T$. Write $\pi_k = \bar{M}_k/h$, where $\bar{M}_k$ is the disclosed numerical representative of $\int_{t_k}^{t_{k+1}} m_0(t)dt$. In the following finite problem, these representatives are fixed real numbers. They specify a feasible, state-

independent reference. The internal transition and transformed reward are

$$Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\sqrt{h} \text{clip}_z(G_k), \quad G_k \sim N(0, I_{d+1}), \quad (21)$$

$$\ell_k(z, a) = B_k \overline{f(z)} + A_k \overline{\log a} - B_k \bar{a} - \frac{\zeta A_k}{2} \bar{a}^2, \quad g_N(z) = -\chi e^{-\rho T} \|\Pi z\|_d^2. \quad (22)$$

Here A_k, B_k are the integrated weights defined above; all discounting is included in them. Let $\mathcal{R}_h(a)$ be the expected sum of these rewards and g_N . The candidate may depend on its entire observed innovation history. Its actions and the reference belong to the same declared convex action box $\mathcal{A}_k$ at each date.

Let V_k^π be the value of this finite problem under π , and define its continuation after choosing the deterministic drift by

$$S_k^\pi(x) = \mathbb{E}[V_{k+1}^\pi\{x + \Sigma\sqrt{h} \text{clip}_z(G_k)\}], \quad q_k^\pi = dD_x S_k^\pi. \quad (23)$$

The argument x is the conditional mean before the current innovation. The current reward ℓ_k is excluded from S_k^π ; subsequent rewards start at $k+1$ on the same global grid. In particular, $q_{N-1}^\pi(x) = -2\chi e^{-\rho T} \Pi x$. Smoothness of f , bounded actions, finite moments, and the quadratic terminal reward justify the derivatives and expectations in this finite problem.

After fitting, freeze an arbitrary continuously differentiable scalar predictor $\hat{S}_k$ and its costate $\hat{q}_k = dD\hat{S}_k$. Set

$$x_b(z) = z + h\{c\mathbf{1} + f(z) - b\}, \quad \hat{Q}_k(z, b) = \ell_k(z, b) + \hat{S}_k\{x_b(z)\}.$$

For U uniform on $[0, 1]$, independent of the candidate's past, the *action bridge* is $X_{k,U} = x_{\pi_k}(Z_k) + U\{x_{a_k}(Z_k) - x_{\pi_k}(Z_k)\}$. Define the following quantities under the candidate's internal-state law:

$$\begin{aligned} \mathcal{M}_h &= \sum_{k=0}^{N-1} \mathbb{E}[\hat{Q}_k(Z_k, a_k) - \hat{Q}_k(Z_k, \pi_k)], \\ \mathcal{A}_h &= \sum_{k=0}^{N-1} h \mathbb{E}\|a_k - \pi_k \mathbf{1}\|_d^2, & \mathcal{E}_h &= \sum_{k=0}^{N-1} h \mathbb{E}\|\hat{q}_k(X_{k,U}) - q_k^\pi(X_{k,U})\|_d^2. \end{aligned} \quad (24)$$

$\mathcal{M}_h$ records the improvement predicted by the fitted scalar continuation, $\mathcal{A}_h$ records the size of the implemented change, and $\mathcal{E}_h$ measures the error that can affect this particular change. Training-state accuracy alone does not identify $\mathcal{E}_h$.

PROPOSITION 2—Finite Bellman bridge: *If the displayed quantities are integrable, then*

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) \geq \mathcal{M}_h - \sqrt{\mathcal{A}_h \mathcal{E}_h}. \quad (25)$$

The assertion holds for every feasible candidate. It requires neither exact maximization of $\hat{Q}$ nor a continuous-time costate approximation. A predictor fitted on another mesh enters $\mathcal{E}_h$ with its full discrepancy from the assessment-grid continuation.

The proposition follows from the finite-horizon performance-difference identity and a line integral of the costate error along each action bridge. It is useful even when action optimization is incomplete. The next result additionally quantifies that incompleteness with a finite, checkable calculation.

PROPOSITION 3—A feasible-actor account: *Suppose $\hat{S}_k$ is twice continuously differentiable and $dD^2\hat{S}_k(x) \preceq \Lambda_k I_d$ everywhere. Let $m_{+,k}$ be the upper endpoint of $\mathcal{A}_k$, and put*

$$\Gamma_k = h^2 \Lambda_k - A_k / m_{+,k}^2, \quad g_{k,i} = A_k \{a_{k,i}^{-1} - \zeta \bar{a}_k\} - B_k - h \hat{q}_{k,i} \{x_{a_k}(Z_k)\}.$$

An upper bound on the actor's $\hat{Q}_k$ maximization gap is

$$\delta_k = \max_{b \in \mathcal{A}_k} \left\{ \langle g_k, b - a_k \rangle_d + \frac{\Gamma_k}{2} \|b - a_k\|_d^2 \right\}. \quad (26)$$

This is a separable scalar calculation. If $\Gamma_k < 0$, its maximizer is the coordinate projection of $a_k - g_k / \Gamma_k$. If $\Gamma_k \geq 0$, one compares the two endpoints in each coordinate. Thus (26) remains valid when global strong concavity of the learned continuation problem cannot be verified.

If $\mu_0 = \min_k (-\Gamma_k / h) > 0$ is verified and $\mathcal{D}_h = \sum_k \mathbb{E} \delta_k$, then

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) \geq \frac{1}{2} \mathcal{M}_h - \mathcal{D}_h - 2\mathcal{E}_h / \mu_0. \quad (27)$$

For exact $\widehat{Q}$ maximizers, $\mathcal{D}_h = 0$ and the coefficient of $\mathcal{E}_h/\mu_0$ improves from two to one. These conclusions apply to the actual feasible action; a small training loss is not substituted for δ_k .

Independent occupation and continuation samples. Draw an independent candidate trajectory, K uniformly from $\{0, \dots, N-1\}$, and U uniformly from $[0, 1]$. At its bridge state, draw two independent continuation banks Z_1, Z_2 with conditional mean $q_K^\pi(X_{K,U})$. A bank may contain antithetic pairs; independence is required between the two banks, not between the members of an antithetic pair. The statistics

$$\begin{aligned} M &= N\{\widehat{Q}_K(Z_K, a_K) - \widehat{Q}_K(Z_K, \pi_K)\}, & A &= T\|a_K - \pi_K \mathbf{1}\|_d^2, \\ E &= T\langle \widehat{q}_K(X_{K,U}) - Z_1, \widehat{q}_K(X_{K,U}) - Z_2 \rangle_d, & D &= N\delta_K \end{aligned} \quad (28)$$

have expectations $\mathcal{M}_h, \mathcal{A}_h, \mathcal{E}_h, \mathcal{D}_h$. Individual values of E may be negative. Replacing them by their positive parts would change the estimand. Moreover, $E^{\text{raw}} = T\|Z_1 - Z_2\|_d^2/2$ estimates the integrated mean-square error of one raw bank on this same occupation law. A confidence bound for $E - E^{\text{raw}}$ therefore assesses the critic's reusable costate prediction against a direct bank with the declared number of rollouts. It concerns this evaluation subproblem; the direct method comparison continues to use the returned policies' independently assessed payoffs.

The supplement gives explicit clipping ranges and expectation allowances for these statistics. They exploit the common quadratic terminal component and antithetic cancellation. They do not use an empirical maximum as a population bound. Independent, equally sized samples for every member of a declared finite stream distribution may be pooled using the maximum proved range and the mean numerical and tail allowances. The resulting target is the uniform mean over all declared streams.

THEOREM 4—From fitted continuation to continuous economic gain: *Condition on the fitted objects and their selection. Suppose a fresh, simultaneous assessment gives $L_M \leq \mathcal{M}_h$, $\mathcal{A}_h \leq U_A$, $\mathcal{E}_h \leq U_E$, and, when used, $\mathcal{D}_h \leq U_D$. Include the disclosed arithmetic and state-representation errors in these endpoints. Define the deterministic holding*

deficit

$$D_{\text{hold}} = \sum_k \left\{ C_k - A_k \log \pi_k + B_k \pi_k + \zeta A_k \pi_k^2 / 2 \right\} \geq 0.$$

For the sampled continuous-economy implementation of Proposition S.10, let $E_{a,h}^{\text{all}}$ be its saved paired payoff-transfer allowance. Then, on that same event,

$$J(a) - J(m_0) \geq L_M - \sqrt{\max(0, U_A) \max(0, U_E)} - D_{\text{hold}} - E_{a,h}^{\text{all}}. \quad (29)$$

When the positive curvature modulus in Proposition 3 is verified, one may also use $L_M/2 - U_D - 2 \max(0, U_E) / \mu_0$ before subtracting the same two allowances. Taking the larger of these two bounds spends no additional probability. Subtracting a valid lower gain bound from the analytical reference regret bound gives a bound against all adapted feasible controls in the original economy.

This account separates three observable economic requirements: a continuation that predicts a useful feasible change, an error small enough under the candidate's occupation law, and a sufficiently accurate implementation. The finite Bellman identity is exact; the transfer of payoffs to the diffusion is quantitative. Neither equality of discrete and continuous costates nor global concavity of the full adapted-control problem is assumed.

4.4. A common continuation for different economic decisions

The capital economy permits a direct study of continuation reuse. Consider a finite-period decision in which a temporary consumption weight, consumption externality, or consumption cap changes at the current date. Subsequent preferences, technology, Gaussian innovations, terminal dispersion cost, and reference consumption remain fixed. For task $\theta = (w, \eta, u)$, the action-dependent payoff is

$$Q_\theta(y, a) = A_0 \{ w \overline{\log a} - \eta \bar{a}^2 / 2 \} - W_0 \bar{a} + S \{ x_a(y) \},$$

$$x_a(y) = y + h \{ c \mathbf{1} + f(y) - a \}, \quad S(x) = \mathbb{E} F(x; \xi), \quad a \in [a, u]^d. \quad (30)$$

Here a known current term independent of a is suppressed. The random variable $F(x; \xi)$ includes the first Gaussian innovation at postdecision mean x , all subsequent

reference rewards, and the terminal payoff. The stored period weights and innovation amplitudes define this finite capital problem. Its decision values have their own interpretation, separate from the continuous-diffusion value in the preceding payoff theorem.

Proposition S.14 establishes that S is identical across these tasks. NBO therefore fits one scalar continuation and queries its gradient when the current economic problem changes. This reuse has a precise information boundary. A stored payoff or costate remains an exact rollout observation at the argument where it was computed. Reusing the same innovation array at another argument requires reevaluating its state recursion and payoff. The Raw comparators may cache valid observations and innovations under the same rule. The comparison concerns the accuracy and complete cost of these representations of the common continuation.

The scalar representation also imposes structure on costate fitting. Fix postdecision arguments and positive weights before drawing rollout labels. Stack the true gradients, multiplied by the square roots of the weights, in q . Let the corresponding labels be $z = q + \varepsilon$, with $\mathbb{E}\varepsilon = 0$ and finite covariance Σ . For differentiable scalar features fixed independently of these labels, let P be the Euclidean orthogonal projection onto their stacked weighted gradient columns. A least-squares scalar critic has fitted gradients Pz , and

$$\mathbb{E}\|Pz - q\|^2 - \mathbb{E}\|z - q\|^2 = \|(I - P)q\|^2 - \text{tr}\{(I - P)\Sigma\}. \quad (31)$$

Proposition S.15 proves the identity without requiring independent coordinates or Gaussian label noise. The variance removed outside the gradient dictionary is the statistical benefit; approximation error is its cost. When the true gradient belongs to the dictionary and the removed variance is positive, the scalar critic strictly reduces this risk. The same fitted object can then serve every task in (30).

This calculation gives a conditional statistical account of the scalar critic. A fully trained nonlinear network generally chooses its features using its labels and need not coincide with this fixed projection. A Raw procedure using the same projection has the same identity. The economic comparisons consequently evaluate all five complete procedures on independent payoff observations, including their

approximation errors, action choices, and computational work. Technical Section S.8 gives the full proofs, direct conditional payoff range, and separate accuracy account over an entire scalar intervention interval.

4.5. Signed continuation assessment

The occupation account can retain the sign of the continuation prediction error. At a sampled candidate state and date, evaluate the reference continuation from the candidate and reference postdecision states using the same future innovations. Add the current reward difference before forming the interval. The conditional expectation is the reference Bellman advantage; averaging over the candidate's occupation distribution gives the finite policy gain.

Proposition S.12 makes this statement precise. The predicted improvement and its signed continuation error cancel sample by sample. A lower endpoint for their combined statistic therefore need not pay separately for the full range of the fitted predictor. Proposition S.13 supplies an economic range using the action tube, bounded production derivatives, and antithetic future paths. It retains the original clipping and arithmetic account and the separate transfer to the continuous-time payoff. The complete derivation is in technical Section S.7.7.

This calculation addresses the precision of an economic identity for the selected policy. The comparison of a learned critic with a Raw continuation remains a distinct empirical question. The repeated-decision design makes that comparison directly, including the cost and benefit of reusing each continuation representation.

5. COMPUTATIONAL DESIGN

The experiments distinguish a decision problem, a complete algorithm, a selected policy, and an independent payoff bank. Each confidence statement conditions on the objects fixed before its bank. The earlier complete method distribution contains sixteen equally weighted algorithm streams at each dimension. Generating new payoff paths for those policies does not create new training configurations. A finite repeated-decision economy likewise has its own value target and is not silently identified with the continuous diffusion.

5.1. *Reusable continuation and new decisions*

The central comparison studies the common continuation in (30) across twelve temporary preference, externality, and consumption-cap tasks. Each task is evaluated at thirty-two fixed states, giving 384 equally weighted queries in each dimension. Four capital calibrations retain the original low-volatility primitives, a two-year quarterly horizon, a four-year half-year horizon, and an intermediate three-year quarterly horizon. The two- and four-year calibrations each have eight decision periods. The new model and its stored period coefficients define a finite repeated-decision economy; no continuous-time transfer is asserted for these menu values.

Five complete procedures share the permitted continuation information: the NBO scalar critic, an unrestricted vector costate predictor, a materialized Raw actor, a direct-policy actor, and query-level Raw sample-average optimization. The vector predictor uses the same postdecision labels as NBO. The Raw actor retains labels at reference actions, whereas direct policy and query-level Raw may retain innovations and reevaluate the continuation when their actions change. Thus the comparisons assess complete representation and optimization procedures, with all approximation and action-selection errors included.

Each calibration and dimension has sixteen streams fixed before confirmation. All methods retain three work stages, using 128, 512, and 2,048 replay states for the fitted procedures and 4, 16, and 64 antithetic paths for query-level Raw. No stage is selected on the final payoff bank. The 216 primary statements cover every method's reference gain and every direct NBO contrast at all stages and cells. Independent confirmation uses 512 antithetic pairs per query and family error 0.018. The fixed materiality and equivalence margins are both 10^{-4} ; a positive interval can also lie inside the equivalence band. A separate family with error 0.002 assesses 128 scalar-intervention accuracy bounds. Technical Section S.19 gives the complete calibration, fitting, inference, and work contracts.

5.2. *Prospective confirmation of previously selected policies*

The confirmation retains all 128 frozen policies from the previous experiment: four methods, sixteen declared streams, and two dimensions. Before generating its new bank, it fixes the original paired-transfer theorem, outward constants, clipping account, reporting code, and an independent family error allocation of 0.01. The new bank has 8,192 paths on 2,048 decision cells. It is independent of the original assessment bank and of the completed training. Its target is the mean payoff of the original finite policy distribution and its specified direct comparisons.

The deterministic transfer is audited separately in technical Section [S.7.6](#). The confirmation retains the original protected constants. It does not substitute a smaller high-precision diagnostic for an interval enclosure. Earlier online stopping decisions remain those made on their original checkpoint banks, irrespective of later confirmation.

5.3. *A strengthened neural-HJB implementation*

This separate continuous-economy family trains new policies at four capital calibrations—high volatility, the original low volatility, a longer horizon, and joint economic stress—in dimensions ten and fifty. Each cell has eight new streams, giving 64 trials, 256 method-group processes, and 320 policy outputs. A new 8,192-path, 2,048-cell bank assesses each trial. All 648 conditional stream and finite-stream-mean events share error allocation 0.01. The stress design changes coupling, terminal dispersion preferences, initial heterogeneity, and the feasible actor radius together; its comparison does not isolate the effect of any one primitive. This family is distinct from both the fixed-policy replication and the sixteen-stream finite menu.

The stronger comparator evaluates the diffusion trace exactly at dimensions ten and fifty. Four learning-rate and collocation configurations are fixed in advance. Each uses Adam followed by L-BFGS; independent residual diagnostics select the configuration and its checkpoint. The final payoff bank does not select the fitted network. The implemented policy records both an optimized action root and a distilled actor, with separately reported payoff and work for each output. Complete work includes every fitted configuration, selection, distillation, action

deployment, and verification. This is a fixed-envelope comparison: a stop based on independent residual checks is not an online economic-target stopping rule. The previous stochastic-trace implementation and its complete outcomes remain in technical Section S.14.

5.4. Precision of the signed occupation account

The signed Bellman experiment uses independent candidate-occupation and continuation banks. Its protected statistic combines the current reward and paired future-return difference before clipping and inference. Its allocation is fixed using the proved economic range and a power calculation. It reports the numerical transfer and every remaining allowance alongside the stochastic radius. The result is interpreted as an assessment of the selected policy’s Bellman account; direct critic–Raw comparisons retain their own payoff, continuation-risk, and work estimands.

5.5. Economic and inferential scope

The confidence family is stated separately for each new experiment. A union of results from different families is asserted only with its explicit allocation. Previous publication evidence retains its original probability account. All results retain their economic parameters, state population, implemented controller, and complete cost boundary. A direct comparison interval that spans the declared margin remains unresolved. A protected lower gain over the schedule, a bound on full-class regret, a finite-grid residual, and an empirical refinement discrepancy answer different economic questions.

6. ECONOMIC VALUE AND COMPUTATIONAL WORK

6.1. The value of a reusable continuation

The complete menu experiment gives a positive economic role for the learned continuation at a stated computational budget. All 640 method fits and 1,920 stage candidates are retained; no fit failed and no stage used the fallback. Table I reports every final-stage direct comparison, pooling the sixteen declared streams and all 384 task-state queries within each of the eight cells. Its units are 10^{-4} , the fixed

materiality and equivalence margin. The intervals belong to the same 216-event family as the complete earlier-stage record.

TABLE I
INCREMENTAL VALUE OF THE REUSABLE NBO CONTINUATION

Economy	d	NBO – Vector	NBO – Raw actor	NBO – DPO	NBO – SAA
Original	10	[0.00, 0.01]	[0.06, 0.07]	[0.06, 0.07]	[-0.01, 0.01]
Original	50	[0.03, 0.05]	[0.25, 0.26]	[0.24, 0.25]	[-0.03, -0.02]
Quarterly	10	[-0.02, 0.02]	[0.20, 0.26]	[0.12, 0.17]	[-0.06, -0.01]
Quarterly	50	[0.22, 0.27]	[0.82, 0.90]	[0.76, 0.83]	[-0.41, -0.36]
Long	10	[0.14, 0.92]	[5.67, 6.70]	[-1.03, -0.19]	[-1.99, -1.22]
Long	50	[16.38, 18.13]	[18.02, 19.72]	[3.27, 4.25]	[-2.68, -1.89]
Intermediate	10	[-0.01, 0.03]	[0.25, 0.33]	[0.15, 0.21]	[-0.03, 0.02]
Intermediate	50	[0.66, 0.79]	[1.47, 1.63]	[1.30, 1.44]	[-0.49, -0.39]

Notes: Intervals are in units of 10^{-4} and use the final, third work stage. Every cell averages all sixteen declared streams and all 384 fixed current-period queries. Each query has 512 independent antithetic future pairs; opposite signs form one observation. Five methods share each pair. All three stages and all eight economic-dimension cells contribute to one 216-event family with error probability 0.018. Current reward and a known terminal quadratic are removed before the random residual is bounded. The point estimator underlying each interval is the clipped residual estimator plus this known offset. Numerical arithmetic, clipping and the unclipped Gaussian law are included. The materiality margin is 10^{-4} .

In both previously unpiloted fifty-dimensional designs, NBO yields materially higher mean payoff than the materialized Raw and direct-policy actors. Its lower gain over the Raw actor is 0.001802 in Long and 0.000147 in Intermediate; the corresponding lower gains over direct policy are 0.000327 and 0.000130, all rounded downward. Each exceeds 10^{-4} . In Long at dimension fifty, NBO also exceeds the fitted vector costate by at least 0.001638. These comparisons measure the economic consequences of using the complete scalar continuation procedure. They do not estimate an independent critic-risk difference or isolate variance removal from nonlinear approximation and optimization.

The payoff improvements over direct policy in those two fifty-dimensional cells accompany lower measured construction and query work. NBO's mean complete three-stage clocks are 14.59 seconds against 38.03 for direct policy in Long, and

TABLE II
COMPLETE CONSTRUCTION AND QUERY WORK

Economy	d	NBO	Vector	Raw actor	DPO	Cached SAA
Original	10	12.52	8.73	6.86	16.21	15.60
Original	50	12.67	9.85	7.07	20.24	27.68
Quarterly	10	10.36	7.69	6.09	21.85	25.23
Quarterly	50	14.34	11.15	8.28	38.93	62.90
Long	10	12.05	8.87	6.96	26.53	30.42
Long	50	14.59	11.06	8.16	38.03	67.04
Intermediate	10	11.96	9.16	7.01	36.55	44.71
Intermediate	50	12.70	9.62	7.18	45.58	79.46

Notes: Seconds per stream, averaged over all sixteen streams, including unsuccessful fits. Each entry is the actual outer-process clock for all three stages, imports, source checks, reusable training data, optimizer and derivative work, all saved queries and durable output. Fixed dependency installation and source extraction precede the clock. Independent confirmation is a shared measured bundle, reported separately in the complete account; the table does not allocate one-fifth of that bundle as an observed standalone verification cost.

12.70 against 45.58 in Intermediate. The materialized Raw actor and vector predictor have lower construction and query clocks than NBO in every final cell. Thus the economic increment over these two representations has a computational cost. The comparison with direct policy instead improves both the protected payoff and the observed work at the stated final budget in the two specified cells.

Cached sample-average optimization supplies a further comparison with direct simulation. In the six final cells outside Long, the complete NBO–SAA interval lies within $[-10^{-4}, 10^{-4}]$. NBO uses less measured construction and query work than the final sixty-four-path SAA in every one of these cells. The reductions range from 19.73 to 84.02 percent of the SAA mean clock. This is a comparison of the prespecified final-stage procedures. It does not identify the minimum work required for a given accuracy: the complete record also retains the cheaper four- and sixteen-path SAA stages.

The Long calibration displays the corresponding accuracy tradeoff. SAA has a protected advantage exceeding 10^{-4} over NBO in both dimensions, at higher final construction and query work. In Long at dimension ten, direct policy also has a

positive protected advantage over NBO, although its material superiority at 10^{-4} is unresolved. Across all three stages, NBO has nine material lower endpoints against Vector, seven against the Raw actor, four against direct policy, and none against SAA, out of twenty-four comparisons with each. The respective numbers of strictly negative NBO upper endpoints are zero, three, seven, and nineteen. The complete intervals in technical Section S.19.5 retain every reversal and every unresolved result. Positive differences and practical equivalence are reported as overlapping conclusions when both inequalities hold.

Independent confirmation is a shared computation. Its complete mean bill ranges from 10.73 to 82.20 seconds per stream across the eight cells; Table S.34 gives every cell and the full clock range. Adding that same bill to each method preserves the final-cell work ordering, but the resulting total is an explicit conservative allocation of the complete five-method, three-stage audit. It is not the time of an independently run single-method verifier. The timing comparisons describe the measured finite collection of runs and carry no unobserved-runtime confidence claim.

TABLE III

CONDITIONAL ACCURACY ON THE PRESCRIBED SCALAR INTERVENTION

Economy	d	Gap $\leq 10^{-4}$	Smallest gap bound	Largest gap bound
Original	10	16/16	0.0000002	0.0000002
Original	50	16/16	0.0000002	0.0000004
Quarterly	10	16/16	0.0000016	0.0000024
Quarterly	50	16/16	0.0000138	0.0000141
Long	10	0/16	0.0440599	0.0454588
Long	50	0/16	0.1962975	0.1963168
Intermediate	10	16/16	0.0000023	0.0000024
Intermediate	50	0/16	0.0002819	0.0002846

Notes: All 128 conditional statements are retained under a separate error probability of 0.002, with 65,536 independent antithetic pairs for each fixed scalar candidate. Each upper bound covers the whole continuous scalar action segment at the predeclared fixed high-dimensional state. It is not a bound on the complete vector action problem or continuous-time adapted policies. Endpoints are rounded upward.

The scalar exercise provides a separate absolute-accuracy result. Eighty of the 128 fixed candidates have a protected gap below 10^{-4} over their entire continuous scalar intervention interval. All sixteen streams attain this criterion in Original and Quarterly at both dimensions and in Intermediate at dimension ten. Long at both dimensions and Intermediate at dimension fifty retain larger gap upper bounds. Nonattainment of an upper certificate does not establish a corresponding lower bound on actual suboptimality. These statements concern the fixed current scalar intervention with the common future reference policy; they do not assert near-optimality over all vector actions or adapted continuous-time controls.

6.2. Independent payoff confirmation

The independent confirmation evaluates all 128 previously selected policies, without retraining or further selection. Table IV reports the schedule-relative method gains and the associated full-adapted-class regret bounds. Table V reports every direct NBO comparison under both the original and the prospectively fixed paired transfer. All eight method endpoints and six direct contrasts belong to the declared 238-event family, with total error allocation 0.01.

The lower endpoints for NBO minus the original neural-HJB implementation are 0.000176 in dimension ten and 0.001342 in dimension fifty, rounded downward. Both exceed the prespecified materiality margin 10^{-4} . In dimension ten the original transfer still gives an interval containing zero on these new observations. Thus the new experiment confirms the value of the sharper paired account under a proof, constants, and reporting rule fixed before its bank. The earlier use of that account remains a post-freeze analytical reanalysis of the earlier observations.

The four NBO–Raw and NBO–direct-policy intervals remain unresolved at 10^{-4} . Their point estimates are close, but the protected intervals do not establish practical equivalence. This confirmation therefore supports the stated comparison with the original HJB implementation and the positive schedule-relative gains; the reusable-continuation experiment supplies the separate comparison of the critic’s incremental economic value. The full-class regret bounds in Table IV retain the anchor gap.

TABLE IV

INDEPENDENT R16 CONFIRMATION OF THE FIXED R15 POLICIES

d	Method	Mean gain	Simultaneous interval	Regret bound
10	NBO	0.001269	[0.000943, 0.001596]	0.025971
10	Raw	0.001268	[0.000942, 0.001595]	0.025972
10	Direct policy	0.001264	[0.000937, 0.001590]	0.025977
10	Neural HJB	0.000974	[0.000646, 0.001302]	0.026268
50	NBO	0.001342	[0.000955, 0.001729]	0.036845
50	Raw	0.001343	[0.000956, 0.001730]	0.036843
50	Direct policy	0.001347	[0.000960, 0.001734]	0.036840
50	Neural HJB	-0.000155	[-0.000560, 0.000249]	0.038359

All 128 original R15 selected policies are evaluated without retraining on new R16 banks. Each of the sixteen original streams receives 8192 independent paths on 2048 cells. The paired theorem, constants and new familywise allocation $\alpha = 0.01$ precede these observations. Interval endpoints and regret upper bounds are rounded outward. This is independent confirmation of fixed policies; historical stopping is unchanged.

TABLE V

PROSPECTIVELY FIXED PAIRED TRANSFER ON FRESH CONFIRMATION BANKS

d	NBO minus	Mean	Original transfer	Fixed paired transfer
10	Raw	0.000001	[-0.000365, 0.000367]	[-0.000114, 0.000115]
10	Direct policy	0.000005	[-0.000361, 0.000372]	[-0.000109, 0.000120]
10	Neural HJB	0.000295	[-0.000076, 0.000667]	[0.000176, 0.000415]
50	Raw	-0.000001	[-0.000432, 0.000429]	[-0.000136, 0.000133]
50	Direct policy	-0.000005	[-0.000436, 0.000426]	[-0.000140, 0.000130]
50	Neural HJB	0.001497	[0.001046, 0.001948]	[0.001342, 0.001653]

All 128 original R15 selected policies are evaluated without retraining on new R16 banks. Each of the sixteen original streams receives 8192 independent paths on 2048 cells. The paired theorem, constants and new familywise allocation $\alpha = 0.01$ precede these observations. Interval endpoints and regret upper bounds are rounded outward. This is independent confirmation of fixed policies; historical stopping is unchanged.

The new per-policy certificates attain the primary schedule-gain target 0.0005 for 14 of 16 NBO policies in dimension ten and all 16 in dimension fifty. The corresponding counts are 13 and 16 for Raw, 10 and 16 for direct policy, and zero in both dimensions for the original HJB implementation. These are outcomes of a new final assessment. They leave the original online stopping decisions unchanged. The pooled dimension-ten HJB lower endpoint exceeds 0.0005, although no individual HJB policy has a new protected lower endpoint exceeding that target. Technical Section S.20 gives the complete target account and additional verification work.

6.3. Strengthened neural-HJB comparison

The strengthened comparison trains each procedure afresh in four continuous diffusion economies, at dimensions ten and fifty, with eight fixed streams per cell. All 64 trials, 256 construction processes, and 320 policy outputs are retained. The exact-trace HJB procedure fits all four declared configurations, chooses its network using independent residual observations, and deploys both the optimized greedy action and its distilled actor. No training failure or reference-policy fallback occurred.

All forty method–economy mean-gain lower endpoints exceed the primary schedule-relative target 0.0005. Table S.43 in the technical supplement gives all gains beside their full-adapted-class regret bounds. These positive policy improvements extend across the declared changes in diffusion risk, horizon, and joint stress. The remaining anchor gap is appreciable, particularly in the longer-horizon and joint-stress economies, so these intervals do not establish 10^{-4} accuracy relative to the adapted optimum.

Table VI reports all 32 direct NBO comparisons. In the low-volatility, fifty-dimensional economy, the lower endpoints against HJB’s greedy and distilled policies are 0.000334 and 0.000340, rounded downward. Both exceed 10^{-4} . The other thirty intervals contain zero and do not establish practical equivalence at that margin. In particular, the comparisons against the strengthened HJB implementation in the original high-volatility economy are unresolved in both dimensions.

This result is distinct from the prospective replication against the original HJB implementation in the preceding subsection.

TABLE VI: Direct common-path NBO comparisons across economic designs

Design	d	NBO minus	Mean	Fixed paired interval
high	10	Raw	-0.000000	[-0.000180, 0.000180]
high	10	Direct policy	0.000004	[-0.000176, 0.000185]
high	10	HJB greedy	0.000008	[-0.000173, 0.000189]
high	10	HJB actor	0.000010	[-0.000171, 0.000191]
high	50	Raw	-0.000002	[-0.000202, 0.000197]
high	50	Direct policy	-0.000001	[-0.000201, 0.000198]
high	50	HJB greedy	-0.000020	[-0.000221, 0.000180]
high	50	HJB actor	0.000040	[-0.000161, 0.000240]
low	10	Raw	0.000000	[-0.000153, 0.000153]
low	10	Direct policy	0.000002	[-0.000151, 0.000155]
low	10	HJB greedy	0.000024	[-0.000129, 0.000177]
low	10	HJB actor	0.000024	[-0.000130, 0.000177]
low	50	Raw	-0.000000	[-0.000174, 0.000174]
low	50	Direct policy	0.000002	[-0.000172, 0.000176]
low	50	HJB greedy	0.000515	[0.000334, 0.000697]
low	50	HJB actor	0.000522	[0.000340, 0.000704]
long	10	Raw	0.000002	[-0.000901, 0.000905]
long	10	Direct policy	0.000020	[-0.000883, 0.000924]
long	10	HJB greedy	0.000060	[-0.000846, 0.000965]
long	10	HJB actor	0.000072	[-0.000834, 0.000977]
long	50	Raw	-0.000003	[-0.001159, 0.001153]
long	50	Direct policy	0.000006	[-0.001151, 0.001162]
long	50	HJB greedy	0.000263	[-0.000897, 0.001423]
long	50	HJB actor	0.000323	[-0.000836, 0.001483]
stress	10	Raw	0.000001	[-0.001038, 0.001040]
stress	10	Direct policy	0.000007	[-0.001033, 0.001047]
stress	10	HJB greedy	0.000021	[-0.001020, 0.001062]

Design	d	NBO minus	Mean	Fixed paired interval
stress	10	HJB actor	0.000038	[-0.001003, 0.001079]
stress	50	Raw	-0.000003	[-0.001254, 0.001247]
stress	50	Direct policy	0.000019	[-0.001233, 0.001271]
stress	50	HJB greedy	-0.000024	[-0.001277, 0.001229]
stress	50	HJB actor	0.000202	[-0.001051, 0.001455]

The continuous-time paired transfer and all calibration-specific constants are fixed before confirmation. Original-transfer intervals remain in the complete machine-readable record. A crossing interval is unresolved, not evidence of equivalence.

The complete fixed-envelope costs also change with the implementation. In the low-volatility, fifty-dimensional cell, mean complete work is 102.27 seconds for NBO, 297.44 for HJB's greedy deployment, and 240.86 for its distilled actor. Raw takes 91.29 seconds there, and its economic comparison with NBO remains unresolved. Table S.44 retains all forty work rows and every final target outcome. Each HJB deployment is charged the common four-candidate construction, selection, distillation, and finalization, together with its own deployment. These are measured fixed-envelope costs; no new online work-to-target stopping experiment is inferred from them.

Technical Section S.21 reports the full economic grid, all four configurations, independent residual diagnostics, trace variance checks, and action diagnostics. Exact derivatives eliminate stochastic trace noise from training the specified network. The held-out diagnostics assess finite collocation banks, and successive value and policy changes compare iterates. Neither quantity is a global value-error certificate. The family contains no direct greedy-minus-distilled payoff event, so the two deployments' separate payoff and work records are not interpreted as a protected distillation-loss comparison. The result concerns these declared procedures and the complete eight-stream populations in each economy.

6.4. A prospective signed occupation account

The new assessment evaluates every original selected NBO policy on fresh occupation paths. Its proof, clipping interval, confidence family, and 1,024 bridges per stream were fixed before the new bank was drawn. The paired continuation calculation uses the same innovations at the candidate and reference action endpoints. It evaluates the signed continuation correction and applies Proposition S.12 directly to their joint Bellman gain, rather than subtracting separately estimated endpoints.

TABLE VII

A FRESH SIGNED BELLMAN MECHANISM ACCOUNT

d	M mean	C mean	L_G	Hold/transfer	Economy lower
10	0.001243	-0.000002	0.001067	0.000300	0.000767
50	0.001354	0.000001	0.001138	0.000363	0.000776

Notes: Each dimension fully enumerates the original sixteen selected NBO streams, with 1,024 independent occupation bridges per stream. One bank of two antithetic pairs supplies the common-noise continuation endpoints. The fitted scalar continuation cancels in $G = M - C$ before inference. All six two-sided (d, M, C, G) statements share a new familywise error probability of 0.01. The finite lower endpoint includes numerical sample and prefix-representation allowances. The final column additionally retains the original holding deficit and full continuous-payoff transfer. This result does not rank the critic against Raw.

In dimension 10, the signed occupation account certifies a continuous-economy gain of at least 0.000767.

In dimension 50, the signed occupation account certifies a continuous-economy gain of at least 0.000776.

The original negative Cauchy–Schwarz mechanism bounds remain part of the record. The present calculation changes the sufficient error account and uses an independently frozen confirmation bank. Its positive endpoints, when obtained, concern the occupation Bellman identity of the returned policy. They do not establish lower prediction risk than Raw or identify a unique incremental contribution of the learned critic. The separate common-data method comparison addresses that question. Both signed lower endpoints exceed the primary target 0.0005; neither attains the secondary target 0.001. The intervals for the signed continuation

correction itself still contain zero. The positive economic conclusion uses the joint Bellman-gain statistic, whose paired future innovations preserve the cancellation in Proposition S.12. The menu comparison measures economic decision values and work. Section 7 supplies the separate independent assessment of continuation-risk differences; the payoff comparisons alone do not imply it.

7. CONTINUATION RISK AT THE ECONOMIC ACTION

The payoff comparison establishes the economic performance of the complete NBO procedure. To assess the fitted continuation itself, we compare its prediction with a cached simulation predictor at the same action pair. For query j , let x_j^+ be the postdecision state of the final NBO action and x_j^0 that of the reference action. The object to be predicted is $c_j = S(x_j^+) - S(x_j^0)$ in the finite economy of Section 4.4. The scalar network supplies $p_{Nj} = \hat{S}_N(x_j^+) - \hat{S}_N(x_j^0)$; cached Raw supplies p_{Rj} . Both are evaluated at these same states. Conditional on the complete training record, these states, and the realized prediction caches, define

$$R_M = J^{-1} \sum_{j=1}^J (p_{Mj} - c_j)^2, \quad M \in \{N, R\}. \quad (32)$$

Conditioning permits nonlinear fitting and learned features. It does not replace the fitted network by a fixed linear dictionary.

A fresh common-future-path contrast Z_{jk} has conditional mean c_j . The difference of its two squared prediction errors is

$$T_{jk} = (p_{Nj} - p_{Rj})(p_{Nj} + p_{Rj} - 2Z_{jk}). \quad (33)$$

Its conditional mean averages to $R_N - R_R$, and its conditional variance is $4(p_{Nj} - p_{Rj})^2 \text{Var}(Z_{jk})$. The common squared-noise term cancels exactly. Centering the remaining continuation residual by its known terminal quadratic component gives a bounded statistic with explicit clipping and arithmetic allowances. Propositions S.26 and S.28 give the complete identities and finite-sample interval. The same scalar differences define a population contrast-training loss (Corollary S.27); the policies reported here are the unchanged fitted policies, not policies retrained with that loss.

The independent assessment fixes all predictions and geometric bounds before drawing its audit innovations. It retains four calibrations, both dimensions, all sixteen streams, and all 384 queries per stream. Each query receives 256 independent antithetic pairs. A sign pair is averaged into one observation; the same audit is reused across the three Raw budgets and is not counted three times as independent information. There are 12,582,912 distinct pair observations over the eight cells. The comparison conditions on each realized Raw cache, rather than averaging over unexecuted caches. It also conditions on NBO's chosen action pairs; it is not an occupation-risk claim for every feasible action or for Raw's separately chosen actor.

TABLE VIII
CONDITIONAL CONTINUATION-RISK DIFFERENCES

Design	d	Raw: 4 paths	Raw: 16 paths	Raw: 64 paths
Low	10	$[-0.0242, 0.00950]$	$[-0.00938, 0.00650]$	$[-0.00491, 0.00486]$
Low	50	$[0.00817, 0.0552]$	$[0.0106, 0.0564]$	$[0.0114, 0.0566]$
Quarterly	10	$[-104, -62.7]$	$[-25.8, -4.52]$	$[-5.94, 8.78]$
Quarterly	50	$[-38.0, -14.9]$	$[-10.2, 3.59]$	$[-2.68, 7.75]$
Long	10	$[-8360, -4760]$	$[-1650, 619]$	$[125, 1900]$
Long	50	$[-5220, -2690]$	$[-1300, 27.5]$	$[-326, 654]$
Intermediate	10	$[-151, -73.0]$	$[-43.7, -4.20]$	$[-12.9, 10.4]$
Intermediate	50	$[-72.2, -15.1]$	$[-20.5, 13.5]$	$[-4.37, 17.2]$

Notes: All intervals are for $R_N - R_R$, in units of 10^{-8} . Negative intervals favor NBO. Endpoints are rounded outward to three significant digits; the complete precision is retained in the report. The 24-event family has simultaneous coverage at least .99. Predictions, action pairs, and realized Raw caches are fixed before the independent assessment.

Table VIII reports all 24 comparisons, not only those favoring learning. Eight intervals have upper endpoint below -10^{-8} and establish lower NBO contrast risk. Four have positive lower endpoint and favor Raw; twelve include zero. At the four-path budget, every Quarterly, Long, and Intermediate cell favors NBO. At sixteen paths, the ten-dimensional Quarterly and Intermediate cells also favor NBO. In contrast, the fifty-dimensional Low cell favors Raw at every budget, and Long in ten dimensions favors the 64-path Raw predictor. This pattern identifies where the

observed fitted continuation improves prediction, and where additional simulation removes that advantage.

The preassigned risk margin 10^{-8} is the square of the payoff unit 10^{-4} , not a payoff difference. Proposition S.25 bounds average binary-decision loss by $\sqrt{R_M}$, but a smaller risk need not give a better action: prediction errors with different signs can reverse that ordering. A difference of risks is not an absolute-risk certificate. Accordingly, the risk result and the actual five-method payoff result provide distinct evidence for NBO; neither is substituted for the other.

The risk family has failure probability at most .01. Combining it with the menu payoff family, whose allocation is .018, gives simultaneous coverage at least .972 by a union bound, without requiring independence between families. Combining it instead with all four earlier scientific families, including their scalar events, gives coverage at least .94. The individual family guarantees remain unchanged. The source freeze, pre-audit receipts, and exact report replay appear in Section S.24.

8. DECISION ACCURACY ACROSS THE COMPLETE BUDGET CATALOGUE

A final-budget comparison alone does not determine which computed procedure meets an economic tolerance at the lowest cost. We therefore use every prespecified menu stage. In each economic cell, the catalogue contains the reference and the five methods at each of three stages. Write V_i for candidate i 's mean payoff over the same complete sixteen-stream population and 384 queries. The target in this section is $g_i = \max_j V_j - V_i$ over these sixteen candidates. This is a catalogue of computed vector decisions, not the full continuous vector-action set or the adapted diffusion class.

The original simultaneous intervals bound five reference-relative gains and four direct NBO contrasts at each stage. If $L_{ab} \leq V_a - V_b \leq U_{ab}$, put directed edges $b \rightarrow a$ of length U_{ab} and $a \rightarrow b$ of length $-L_{ab}$. Let D_{ij} be the shortest-path length from i to j . The same confidence event then gives

$$\max_j \{-D_{ji}\} \leq g_i \leq \bar{g}_i := \max_j D_{ij}. \quad (34)$$

Proposition S.29 proves that these bounds are sharp in the uncertainty set defined by the supplied intervals. No new simulation, normal approximation, or additional

probability allocation is used. For a tolerance τ , let $\mathcal{C}_\tau = \{i : \bar{g}_i \leq \tau\}$. Selecting the least recorded cost within $\mathcal{C}_\tau$ preserves the simultaneous accuracy guarantee. It identifies the cheapest *certified* candidate, not necessarily the cheapest candidate whose unknown true regret is below τ .

TABLE IX

LEAST ACCOUNTED COST WITH A CATALOGUE ACCURACY CERTIFICATE

Design	d	Certified choice	$\bar{g}$	Seconds	NBO (3): $\bar{g}$
Low	10	SAA (1)	0.03935	3.47	0.03883
Low	50	SAA (1)	0.03563	3.97	0.05945
Quarterly	10	Raw actor (2)	0.7543	4.85	0.1848
Quarterly	50	Vector (3)	0.7678	11.15	0.5235
Long	10	SAA (3)	0	30.42	1.984
Long	50	SAA (3)	0.4893	67.04	2.688
Intermediate	10	Vector (2)	0.8382	6.52	0.4282
Intermediate	50	NBO (3)	0.9140	12.70	0.9140

Notes: Regret bounds are in units of 10^{-4} ; the target is at most one. Parentheses identify the original stage. The catalogue is the reference plus fifteen method-stage procedures, not the full action space. Early-stage seconds are prefix allocations; stage-three seconds are complete construction/query clocks. Shared assessment and the complete comparative bill are reported separately. The selected cost is least among certified candidates only.

The fifty-dimensional Intermediate design gives a positive NBO accuracy–work result against the complete budget catalogue. NBO’s final-stage regret upper bound is 0.000091391 (rounded upward), below 10^{-4} , at mean construction and query cost 12.6993 seconds (rounded upward). It is the least-cost candidate with this certificate. The cheaper final Raw and vector actors have regret lower bounds above 0.0001874 and 0.0001064, respectively. The cheaper four-path SAA and second-stage NBO candidates remain unresolved: their regret intervals straddle 10^{-4} . Thus the certified choice does not establish actual minimum work over these unresolved alternatives.

Table IX retains the choices in every other design. The final NBO candidate is within 10^{-4} of the best catalogue payoff in six cells. Both Long cells instead

select final-stage SAA; elsewhere a cheaper Raw actor, vector predictor, or low-budget SAA has the certificate. The lower-cost alternatives are part of the result, not exclusions from the comparison.

Early-stage costs are the original construction-prefix allocations, not independent clocks for a prospectively stopped procedure. Final-stage costs are measured complete three-stage construction and query processes. The shared confirmation process is charged separately and in full; adding this same bill to each candidate does not alter its ordering. The total cost of having conducted the comparison includes *all* five construction processes and that shared assessment, not just the selected candidate. Section S.24 reports these complete bills and every candidate. The closure and resulting selection are a post-freeze deterministic reanalysis of an already simultaneous family, not a newly executed stopping rule or a fresh confirmatory experiment.

8.1. *The previous evidence and its chronology*

The previous complete experiment used higher volatility in the same capital model, sixteen declared streams per method and dimension, and a common independent final payoff assessment. The complete protocols, records, and endpoint tables remain current in technical Sections S.14 and S.15. The following facts determine how the new results are compared with that evidence.

At dimension ten, the original NBO-minus-HJB interval was approximately $[-0.000073, 0.000663]$. It did not establish superiority. After the original analytical freeze and inspection of interval widths, the common-innovation transfer was refined on the same observations. Its resulting interval was $[0.000179, 0.000411]$. At dimension fifty the original interval was $[0.001051, 0.001945]$ and the refined interval was $[0.001347, 0.001650]$. Endpoints in this paragraph are rounded outward. The refinement changes a deterministic allowance, not the simulated policies, observations, original confidence event, or earlier stopping decisions. It is labelled a post-freeze analytical reanalysis throughout the present comparison.

All four refined NBO–Raw and NBO–DPO direct intervals remained unresolved at the 10^{-4} margin. They established neither a ranking nor practical equivalence.

At dimension fifty, all sixteen NBO, Raw, and DPO streams reached the original online gain target 0.0005. Their mean complete process times were approximately 357.80, 346.22, and 596.94 seconds. The specified HJB implementation reached that online target in no stream. At dimension ten no method reached the online target; the later final certificates do not change that outcome. A pooled dimension-ten HJB lower gain above 0.0005 is a method-mean statement even though no individual HJB stream had a final certificate at that target.

The original sufficient mechanism lower bounds were approximately -0.006778 and -0.004800 . Its critic-Raw continuation-risk contrasts were also unresolved. The NBO full-adapted-class regret upper bounds, rounded upward, were 0.025971 and 0.036843. These are absolute accuracy bounds with their own anchor gap, rather than a 10^{-4} near-optimality certificate. The new signed account and strengthened comparison are assessed alongside these retained outcomes.

The scalar and observation experiments provide separate information. The scalar Howard calculation reports residual, restriction, mesh, and domain accounts for its finite-grid problem. Its deployed classical policies outperform the frozen historical neural policies in the measured scalar comparisons, and the scalar NBO-DPO ranking reverses at one initial state. In the observation experiment all four protected parent comparisons have positive lower gains, while six of twelve measurement specifications retain a positive continuous-economy lower gain after sensing allowances. Each statement remains tied to its own model parameters, policy, decision grid, and information structure.

8.2. *Information and the implemented decision rule*

The economic institution is a controller that receives capital observations at specified dates, stores an internal state, and holds the selected consumption share until the next decision date. Its economic primitive is the original capital diffusion. Its internal numerical state need not equal the physical state. When the controller uses private randomization, that randomization is part of its specified information and is included in the law of the evaluated policy.

The exact-history representation in Proposition S.18 assumes continuous noise-less observation of every capital coordinate, known coefficients and actions, exact drift integration, and an independent private Brownian motion. Under those assumptions the observation history supplies a Brownian representation with the law required by the innovation-driven controller. It does not recover the original latent null-space shock path by path. Proposition S.19 states why the added independent randomization leaves the economic optimum unchanged. These are representation results for a randomized history-dependent policy.

Finite observations require a different implementation account. Proposition S.20 bounds the difference between the measured-state controller and its specified protected parent, retaining sensor error, the internal recurrence, action-map error, clipping, and arithmetic. Its allowance belongs to that parent and that time grid. The empirical comparison reports it alongside the gain measured for the same parent and population; a certificate for a different mesh is not substituted. Complete assumptions and proofs are in Appendix S.10.

8.3. When is state-dependent management worth its fee?

To give the policy differences a decision interpretation, consider a proportional resource cost $\tau \in [0, 1)$ for administering a state-dependent consumption rule. Gross withdrawals from capital remain $m_i e^{Y_i}$, and the service absorbs $\tau m_i e^{Y_i}$. Household consumption is $(1 - \tau) m_i e^{Y_i}$. Production, gross depletion, the gross-withdrawal adjustment cost, and terminal capital are unchanged. This is a self-financed resource experiment rather than an externally financed utility supplement. It changes the specified fee parameter, not the model used to train or benchmark the gross policy.

Let $A_T = \int_0^T e^{-\rho t} dt$. Log felicity gives the exact identity

$$J_{\mathcal{P}}^{\tau}(a) - J_{\mathcal{P}}(m_0) = J_{\mathcal{P}}(a) - J_{\mathcal{P}}(m_0) + A_T \log(1 - \tau). \quad (35)$$

Hence a positive certified improvement L_a supports every fee τ satisfying

$$\tau < 1 - \exp(-L_a / A_T).$$

A negative upper endpoint after the fee rules out an improvement for that candidate under the confidence event; an interval spanning zero is inconclusive. The fee grid is fixed before the final simulation. We do not equate wall-clock cost to a monetary fee or call these deliberately stylized balance sheets a calibrated household population. The experiment instead states precisely which adoption decisions are supported by the verified utility differences.

9. ECONOMIC APPLICATIONS OF THE BELLMAN STRUCTURE

The motivating economic problem combines external choices with the costly adjustment of preferences. Recursive utility changes how a given policy is evaluated; preference adjustment adds an internal choice to its Hamiltonian; and strategic interaction changes whose deviations define improvement. These distinctions determine the economic interpretation of NBO. The capital economy studied above provides a setting in which policy-specific verification can be carried through to a high-dimensional diffusion. The preference, temporal-self, and game applications establish the corresponding economic objects and their own verification results.

9.1. *Costly preference adjustment and portfolio choice*

Let u denote risk aversion and X financial wealth. The household chooses consumption c , portfolio share p , and preference-adjustment effort θ . The preference state is reflected on a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, and wealth follows the portfolio diffusion:

$$\begin{aligned} du &= \theta dt + \sigma_u dW^u + dL^u - dL^{\bar{u}}, \\ dX &= \{[r + p(\mu - r)]X - c\} dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \end{aligned} \quad (36)$$

For a fixed consumption unit $\bar{c} > 0$, flow utility is

$$U(c, u) - \frac{1}{2}k\theta^2, \quad U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}.$$

The objective discounts both felicity and adjustment costs at rate ρ and includes the specified wealth-exit and terminal payoff. The baseline uses $\bar{c} = 1$. Because the household controls u , the cardinal normalization across preference states is a

primitive: adding a function of u changes the incentives to adjust preferences. The reflected state, positive consumption domain, and stopping data give the evaluation equation its economic and mathematical domain.

At an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the three policy conditions are

$$\begin{aligned} c^* &= \bar{c}(\bar{c}V_X)^{-1/u}, & \theta^* &= V_u/k, \\ p^* &= -\frac{V_X(\mu - r)}{XV_{XX}\sigma_X^2} - \frac{\rho\sigma_u V_{uX}}{XV_{XX}\sigma_X}. \end{aligned} \quad (37)$$

Box constraints require their corresponding feasible maximizers. The cross derivative in the portfolio rule links the external risk choice to the internal preference state. Its sign is an equilibrium object to be determined by evaluation. The adjustment rule likewise relates effort to the shadow value of changing preferences. A higher adjustment cost lowers the value of every fixed policy and hence the optimized value; whenever an envelope derivative exists,

$$\partial_k V^k(t, u, X) = -\frac{1}{2} \mathbb{E}_{t,u,X}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (38)$$

This comparative static holds with the feasible class and stopping payoff fixed.

The numerical application implements these choices jointly. The finite-action study evaluates complete policies by independent backward recursion. The continuous-action study trains the actor through its own payoff and verifies its hybrid policy with a complete action-box enclosure. Fresh training on nested grids separates the effect of resolution from the evaluation of a frozen coarse policy. A model-specific compensating-consumption calculation translates its cardinal utility bounds into an explicit transfer experiment. Application Sections [A.1.3](#), [A.1.3.4](#), and [A.1.8](#) give the full model, all recorded policies, refinement evidence, and compensation account.

9.2. Recursive utility and temporal selves

Recursive utility enters policy evaluation through the policy's own continuation utility. With $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, the Epstein–Zin aggregator used

here is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (39)$$

The invariant utility interval and boundary payoff are part of the evaluation problem. Application Section A.1.2 derives the homothetic policy conditions and verifies the matching-bequest experiment, including the barriers that keep every policy utility in the aggregator's domain. This application tests recursive evaluation while maintaining the same separation between value approximation and feasible action improvement.

Time inconsistency requires a different improvement criterion. For the continuous discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta) e^{-(\rho + \kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0,$$

two critics evaluate the same future policy at the two discount rates. Writing these values as W_1, W_2 , the current self uses $V = \beta W_1 + (1 - \beta) W_2$ and chooses an instantaneous deviation by maximizing $u(a) + \mathcal{L}^a V$. Application Section A.1.4 states the complete component equations and derives the spike-deviation condition. In the homothetic consumption economy, the equilibrium consumption share is the unique root in its admissible interval, and stronger present bias raises that share. Independently optimized finite-duration deviations check the limiting equilibrium condition. Thus the actor's objective follows from the temporal equilibrium concept, while the critics retain their policy-evaluation role.

9.3. Capital investment and strategic deviations

In the dynamic Cournot application, firm i chooses investment in its own capital, with

$$dK^i = (I^i - \delta K^i) dt + \sigma_K K^i dW^i, \quad q_i = AK^i.$$

Its flow profit is $P_N(K)q_i - I^i - b_i(I^i)^2/2$, where the market-size normalization in P_N is specified for each economy. Given rivals' policies, a critic evaluates the resulting own-firm payoff and the actor changes only its own action. Consequently,

the relevant economic error is the gain from a complete unilateral deviation against the frozen rival profile.

Application Sections A.1.5 and A.1.5.3 specify the boundaries, investment sets, and market normalizations, and report independent full dynamic best responses to every principal finite neural profile. The resulting profit-unit bounds include all dates and states of the stated finite game. They also cover randomized and history-dependent unilateral deviations against those fixed Markov rivals, by finite dynamic programming. The record reports the cost of the exhaustive training oracle and any nonzero one-step equilibrium defects along with the final exploitability calculation.

These applications give the common method a precise economic interpretation. A recursive critic evaluates a utility process; preference and capital actors allocate resources and adjustment effort; a temporal actor satisfies a spike condition; and a strategic actor is assessed against unilateral best responses. Each application supplies the boundary conditions, approximation class, and error account required by its own economic question.

10. CONCLUSION

Neural Bellman Operators connect a specified continuation object with feasible economic decisions. The nonlinear capital economy makes this connection explicit: an exact finite Bellman identity accounts for continuation error along the chosen action change, while a separate diffusion account assesses the deployed continuous-time policy. Repeated decisions permit the same continuation representation to be used across economic tasks. A fixed gradient-dictionary calculation identifies the approximation–variance tradeoff of a linear projection, and the numerical experiment measures the economic performance of the nonlinear fitted procedure.

The complete continuation-menu experiment establishes material NBO payoff gains over materialized Raw and direct-policy actors in two previously unpiloted fifty-dimensional economies. NBO also uses less construction and query work than direct policy in those cells. At the final prescribed SAA budget, six of eight economic cells have equivalent payoffs within 10^{-4} and lower construction and query work for NBO. Cached SAA has materially higher payoffs in both longest-

horizon cells. The full set of earlier budgets, cheaper vector and Raw actors, and common confirmation costs remains part of the evidence. This result identifies specific economic gains from a reusable fitted continuation and the work needed to obtain them.

Independent confirmation supports the original-policy paired comparison under a prospectively fixed analysis. The strengthened HJB procedure shows why the implementation must remain explicit: NBO's material advantage over both new HJB deployments is established in the low-volatility, fifty-dimensional economy, while the remaining direct comparisons are unresolved. A new signed continuation statistic verifies positive occupation Bellman gains by retaining the common-path cancellation. These positive payoff statements coexist with the reported full-class regret bounds and the finite menu's separate scalar accuracy account.

The continuation-risk comparison supplies separate evidence about the fitted nonlinear object. Cancellation of common simulation noise makes eight risk advantages and four reversals identifiable across the complete 24-event family. The economic payoff experiments, rather than risk ordering alone, establish the value of the returned decisions. Exact closure of the complete menu intervals further identifies the least-cost certified candidate at the common accuracy tolerance. NBO attains that distinction in the fifty-dimensional Intermediate design; other designs select simulation or cheaper fitted actors. The result concerns an explicit catalogue of computed procedures and observed accounting costs. It does not replace the broader adapted-class regret account or the full bill for producing the comparison.

The economic applications retain their own utility domains, boundary conditions, and deviation criteria. These determine the continuation that must be learned and the decision that constitutes an improvement. The current application companion preserves the complete models, derivations, proofs, and numerical records for recursive utility, endogenous preferences, temporal selves, and strategic interaction. The common evaluation-improvement structure makes their computational requirements comparable while retaining their distinct economic content.

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